

Student: Jonatan Rabjerg
Date: 1/15/24

Instructor: Andreas Basse-O'Connor
Course: Calculus Beta

Assignment: MyLab Math - Uge 11

Solve the given equation by the method of undetermined coefficients.

$$y'' - 7y' + 6y = 18$$

The general solution to a nonhomogeneous differential equation $ay'' + by' + cy = G(x)$ is the sum of y_c and y_p , where y_c is the general solution to the complementary homogeneous equation $ay'' + by' + cy = 0$ and y_p is any particular solution to the nonhomogeneous equation.

$$y = y_c + y_p$$

First, find y_c by solving the homogeneous differential equation $y'' - 7y' + 6y = 0$. The auxiliary equation for $y'' - 7y' + 6y = 0$ is shown below.

$$r^2 - 7r + 6 = 0$$

Solve the auxiliary equation.

$$(r - 6)(r - 1) = 0$$

$$r = 6 \text{ or } r = 1$$

The auxiliary equation has two real and unequal roots. The general solution to an equation of this type is of the form $y = c_1 e^{r_1 x} + c_2 e^{r_2 x}$.

Substitute the values of r_1 and r_2 into this form to find the general solution y_c for the complementary homogeneous equation $y'' - 7y' + 6y = 0$.

$$y_c = c_1 e^{6x} + c_2 e^x$$

Now, find y_p , a particular solution to the given nonhomogeneous differential equation.

If $G(x)$ is a polynomial, y_p has the form of a polynomial of the same order as $G(x)$. In this case, $G(x)$ is a constant. Therefore, y_p must also be a constant.

$$y_p = C$$

To find the value of C , substitute y_p and its derivatives into the given nonhomogeneous differential equation. Start by finding the first and second derivatives of y_p .

$$y_p' = 0$$

$$y_p'' = 0$$

Substitute y_p and its derivatives into the given nonhomogeneous equation and solve for C .

$$0 - 7(0) + 6C = 18$$

$$6C = 18$$

$$C = 3$$

Therefore, the particular solution y_p is 3. Add y_p to y_c to obtain the general solution to $y'' - 7y' + 6y = 18$.

$$y = y_c + y_p$$

$$y = c_1 e^{6x} + c_2 e^x + 3$$